

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 02: CELL: THE BUILDING BLOCK OF LIFE

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: TEMPORARY MOUNT OF ONION PEEL & MICROSCOPIC CYTOLOGY

Students prepared a temporary mount of an onion bulb leaf peel to examine plant cell architecture under a compound microscope.

1. They peeled a thin translucent layer from the concave inner surface of an onion scale leaf.
2. The peel was transferred to a watch glass containing water to prevent desiccation.
3. It was stained with a dilute solution of Safranin for 2 minutes and washed.
4. The stained peel was mounted on a clean glass slide in a drop of glycerin and covered with a glass coverslip, gently lowering it with a needle to avoid trapping air bubbles.

Under 100x and 400x magnification, they observed contiguous rectangular cells with distinct boundaries.

Q.1 Why is glycerin used as a mounting medium instead of plain water?

- | | |
|---|---|
| (A) Glycerin dissolves the cell wall to reveal organelles | (B) Glycerin prevents the biological specimen from drying out during extended microscopic observation |
| (C) Glycerin stains the chromosomes bright purple | (D) Glycerin acts as an adhesive to permanently glue the coverslip |

Q.2 What is the primary biological purpose of staining the onion peel with Safranin?

- | | |
|---|--|
| (A) To kill bacteria on the slide | (B) To impart contrast by staining cell walls and nuclei, making cellular structures clearly visible |
| (C) To induce active mitosis in the cells | (D) To make the cell membrane permeable to glucose |

Q.3 Which organelle is prominently observed as a dense, darkly stained spherical body located towards the periphery of each onion cell?

- | | |
|-------------------|-----------------|
| (A) Mitochondrion | (B) Chloroplast |
| (C) Nucleus | (D) Centriole |

Q.4 Why are chloroplasts absent in the cells of an onion scale leaf peel?

- (A) Onion is an animal organism (B) Onion scale leaves are underground storage organs and are not exposed to sunlight for photosynthesis
- (C) The cells were killed by Safranin staining (D) Chloroplasts were filtered out during peeling

Q.5 What visual artifact indicates that the coverslip was dropped improperly onto the mount?

- (A) Circular dark-bordered rings that move when the slide is pressed (air bubbles) (B) Elongated rectangular cell walls
- (C) Bright red nuclei (D) Clear transparent cytoplasm

CASE STUDY 2: PLASMOLYSIS & OSMOTIC MECHANICS IN *RHOEO DISCOLOR* LEAF

A biology teacher instructed students to investigate osmosis and plasmolysis using the colored lower epidermal peel of *Rhoeo discolor* leaf.

- The lower epidermis of *Rhoeo* contains anthocyanin pigment dissolved in its vacuolar sap, giving cells a striking purple color.
- **Slide 1:** A fresh peel mounted in pure water.
- **Slide 2:** A fresh peel mounted in concentrated (20%) sucrose solution and observed after 5 minutes.
- **Slide 3:** A leaf peel boiled in water for 5 minutes, then mounted in concentrated (20%) sucrose solution.

Q.6 What morphological change is observed in the epidermal cells of Slide 2 (mounted in 20% sucrose)?

- (A) The cells swell up and burst (B) The purple protoplast shrinks away from the cell wall (plasmolysis) as water exits the vacuole via exosmosis
- (C) The cells double in size due to endosmosis (D) The cell wall dissolves completely

Q.7 What occupies the clear space between the shrunken protoplast and the cell wall in a plasmolysed plant cell?

- (A) A total vacuum (B) Pure air
- (C) The external 20% sucrose solution (D) Concentrated cytoplasm

Q.8 What happens when pure water is added to the plasmolysed cells of Slide 2 via irrigation?

- (A) The cells undergo permanent apoptosis (B) Water enters via endosmosis, restoring protoplast turgidity against the cell wall (deplasmolysis)
- (C) The cells burst instantly (D) The purple anthocyanin pigment precipitates into crystals

Q.9 What was observed on Slide 3 (boiled leaf peel in 20% sucrose solution)?

- (A) Rapid violent plasmolysis (B) Zero plasmolysis; cells remained unchanged because boiling destroyed the living semi-permeable plasma membrane
- (C) The cells divided into two daughter cells (D) The purple color turned bright green

Q.10 Which physical pressure is exerted by the swollen protoplast outward against the cell wall in a fully turgid plant cell?

- (A) Suction pressure (B) Turgor pressure
- (C) Atmospheric pressure (D) Capillary pressure

CASE STUDY 3: OSMOTIC FRAGILITY IN ANIMAL CELLS (ERYTHROCYTES)

In a physiology experiment, mammalian red blood cells (RBCs) were suspended in test tubes containing sodium chloride (\$NaCl\$) solutions of varying molar concentrations at 37°C:

- **Tube A:** 0.9% NaCl solution (Isotonic saline)
- **Tube B:** 0.2% NaCl solution (Hypotonic)
- **Tube C:** 2.0% NaCl solution (Hypertonic)

After 15 minutes, samples from each tube were examined under a light microscope and their optical absorbance was measured.

Q.11 What is the physical appearance of the red blood cells in Tube A (0.9% saline)?

- | | |
|---------------------------|--|
| (A) Swollen and spherical | (B) Normal biconcave disc shape with zero net water movement |
| (C) Shriveled and spiky | (D) Completely lysed into fragments |

Q.12 What happens to the red blood cells placed in Tube B (0.2% NaCl hypotonic solution)?

- | | |
|--------------------------------------|---|
| (A) They lose water and crenate | (B) Water rapidly enters via endosmosis, causing cells to swell and burst (hemolysis) |
| (C) They form a rigid cellulose wall | (D) They divide via binary fission |

Q.13 What morphological term describes the shriveled, scalloped appearance of RBCs in Tube C (2.0% NaCl)?

- | | |
|-----------------|--------------------------|
| (A) Turgid | (B) Crenated (Crenation) |
| (C) Plasmolysed | (D) Hypertrophied |

Q.14 Why do plant cells not burst when subjected to the same hypotonic conditions as Tube B?

- | | |
|--|--|
| (A) Plant cells lack a plasma membrane | (B) The rigid plant cell wall exerts an equal counter wall pressure that opposes increasing internal turgor pressure |
| (C) Plant cells do not absorb water | (D) Plant cell sap contains no solutes |

Q.15 Which clinical medical practice directly relies on understanding isotonic saline concentrations?

- | | |
|--|---|
| (A) Injecting pure distilled water directly into patient veins | (B) Administering intravenous (IV) fluid drips matching 0.9% saline or 5% dextrose osmolarity |
| (C) Prescribing concentrated sea water for dehydration | (D) Boiling blood before transfusions |

CASE STUDY 4: CELL FRACTIONATION & ORGANELLE BIOENERGETICS

Biochemists homogenized fresh liver tissue in chilled isotonic buffer and subjected the cellular suspension to differential centrifugation at increasing centrifugal speeds:

- **Pellet 1 (Low speed, 1,000 g):** Nuclei and unbroken cells sedimented.
- **Pellet 2 (Medium speed, 10,000 g):** Mitochondria, lysosomes, and peroxisomes sedimented.
- **Pellet 3 (High speed, 100,000 g):** Microsomes (fragments of ER and Golgi membranes) sedimented.
- **Supernatant:** Soluble cytosol containing ribosomes and enzymes.

Upon analyzing Pellet 2, researchers observed high rates of ATP synthesis in the presence of oxygen, pyruvate, and ADP.

Q.16 Which organelle in Pellet 2 is responsible for generating ATP through cellular aerobic respiration?

- | | |
|----------------|-------------------|
| (A) Lysosome | (B) Mitochondrion |
| (C) Peroxisome | (D) Centrosome |

Q.17 What morphological adaptation of the inner mitochondrial membrane enhances its ATP production efficiency?

- (A) Pores made of cellulose fibers (B) Extensive inward foldings called cristae that substantially increase surface area for respiratory enzymes
- (C) A thick layer of hydrophobic wax (D) Complete absence of membrane proteins

Q.18 Which unique genetic feature allows mitochondria to synthesize some of their own proteins?

- (A) They possess their own circular DNA and 70S ribosomes (B) They borrow mRNA from lysosomes
- (C) They have nuclear pores on their outer membrane (D) They contain linear chromosomes with histones

Q.19 If the lysosomes present in Pellet 2 are subjected to detergent lysis, which class of enzymes will be released?

- (A) Photosynthetic carboxylases (B) Acid hydrolytic digestive enzymes (proteases, nucleases, lipases)
- (C) DNA polymerases active at pH 9 (D) Chlorophyll synthesis enzymes

Q.20 Why was the tissue homogenization performed in a chilled, isotonic sucrose buffer rather than in pure warm water?

- (A) To prevent osmotic lysis of organelles and inhibit degradative enzymes through low temperature (B) To speed up chemical reactions
- (C) To dissolve the mitochondrial membranes (D) To boil away unwanted water

CASE STUDY 5: ENDOMEMBRANE SECRETORY TRANSPORT & PROTEIN TARGETING

In a pulse-chase experiment, pancreatic acinar cells synthesizing digestive enzymes were briefly exposed ('pulse') to radioactively labeled amino acids (^3H -leucine) for 3 minutes, then incubated ('chase') with non-radioactive amino acids. Samples were fixed at intervals and examined by autoradiography under an electron microscope to track the intracellular route of the synthesized digestive enzymes:

- At 3 minutes: Radioactivity concentrated exclusively on the Rough Endoplasmic Reticulum (RER).
- At 20 minutes: Radioactivity shifted to the Golgi apparatus.
- At 90 minutes: Radioactivity accumulated in secretory vesicles near the apical plasma membrane.

Q.21 Which organelle component attached to the cytosolic surface of the RER synthesized the radioactive proteins?

- (A) Ribosomes (B) Lipid droplets
- (C) Centrioles (D) Microfilaments

Q.22 What is the primary processing function of the Golgi apparatus as proteins pass through its flattened cisternae?

- (A) Replication of genomic DNA (B) Modification, glycosylation, packaging, and sorting of proteins into membrane-bound vesicles
- (C) Direct synthesis of ATP via oxidative phosphorylation (D) Digestion of glucose into pyruvate

Q.23 How does the Smooth Endoplasmic Reticulum (SER) differ biochemically and functionally from the RER?

- (A) SER contains ribosomes and makes RNA (B) SER lacks ribosomes and is specialized for lipid and steroid synthesis, as well as drug detoxification
- (C) SER is located inside the nucleus (D) SER performs cellular photosynthesis

Q.24 By which transport mechanism do mature secretory vesicles release digestive enzymes across the cell's outer membrane into pancreatic ducts?

- (A) Phagocytosis (B) Exocytosis
- (C) Endosmosis (D) Simple diffusion

Q.25 Which organelle would degrade dysfunctional, worn-out Golgi or ER fragments inside the cell?

(A) Mitochondria

(B) Lysosomes (via autophagy)

(C) Chloroplasts

(D) Centrioles

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)

02 (A) (B) (C) (D)

03 (A) (B) (C) (D)

04 (A) (B) (C) (D)

05 (A) (B) (C) (D)

06 (A) (B) (C) (D)

07 (A) (B) (C) (D)

08 (A) (B) (C) (D)

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10 (A) (B) (C) (D)

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15 (A) (B) (C) (D)

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19 (A) (B) (C) (D)

20 (A) (B) (C) (D)

21 (A) (B) (C) (D)

22 (A) (B) (C) (D)

23 (A) (B) (C) (D)

24 (A) (B) (C) (D)

25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	Glycerin is hygroscopic and has high optical clarity; it prevents dehydration and shrinkage of the specimen under the warm microscope lamp.
Q.2	(B)	Unstained living plant cells are mostly transparent. Safranin stains lignified and cellulosic cell walls and chromatin pink/red, providing visual contrast.
Q.3	(C)	The nucleus stains darkly with Safranin and is pushed towards the peripheral cytoplasm due to the presence of a large central vacuole.
Q.4	(B)	Onion bulbs grow underground to store food. In the absence of sunlight, plastids differentiate into colorless leucoplasts rather than green chloroplasts.
Q.5	(A)	Air bubbles trapped under the coverslip appear as spherical or oval structures with thick, dark, refractive boundaries that interfere with observation.
Q.6	(B)	The 20% sucrose solution is hypertonic to the cell sap. Water moves out via exosmosis, causing the protoplast to shrink away from the rigid cell wall (plasmolysis).
Q.7	(C)	The plant cell wall is freely permeable to water and dissolved sucrose, whereas the plasma membrane is semi-permeable. Hence, the hypertonic sucrose solution fills the gap.
Q.8	(B)	If placed back into a hypotonic medium (water) before dying, water rushes back in via endosmosis, reversing plasmolysis (deplasmolysis).
Q.9	(B)	Boiling denatures membrane transport proteins and disrupts the phospholipid bilayer, killing the cells. Dead membranes cannot carry out selective osmosis.
Q.10	(B)	Turgor pressure is the hydrostatic pressure exerted by the fluid contents of the vacuole and cytoplasm against the plasma membrane and cell wall.
Q.11	(B)	0.9% NaCl is physiologically isotonic with mammalian intracellular cytoplasm; dynamic equilibrium exists with zero net gain or loss of water.
Q.12	(B)	Animal cells lack a rigid cell wall. In a hypotonic medium, endosmosis increases internal pressure until the plasma membrane ruptures, releasing hemoglobin (hemolysis).
Q.13	(B)	In animal cells, loss of water to a hypertonic medium causes cellular shrinkage and notched, spiky margins, termed crenation.
Q.14	(B)	The tough cellulose wall exerts inward wall pressure (\$WPS\$) equal to outward turgor pressure (\$TPS\$), preventing osmotic bursting.
Q.15	(B)	IV fluids must be strictly isotonic to human blood plasma (e.g., 0.9% normal saline) to prevent fatal hemolysis or crenation of red blood cells in vivo.
Q.16	(B)	Mitochondria are the powerhouses of eukaryotic cells, coupling electron transport and oxidative phosphorylation to synthesize ATP.
Q.17	(B)	Cristae greatly amplify the total surface area available for electron transport chain complexes and ATP synthase enzymes.
Q.18	(A)	Mitochondria are semi-autonomous endosymbionts containing circular prokaryotic-like DNA and 70S ribosomes.
Q.19	(B)	Lysosomes are acidic hydrolytic vesicles containing enzymes that digest proteins, nucleic acids, lipids, and carbohydrates.
Q.20	(A)	Isotonic buffer prevents organelles from swelling and bursting osmotically, while chilling (4°C) minimizes destructive enzymatic degradation.
Q.21	(A)	Ribosomes attached to the outer surface of the rough endoplasmic reticulum translate mRNA into secretory and membrane proteins.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.22	(B)	The Golgi apparatus modifies proteins (e.g., adding carbohydrates), packages them into vesicles, and directs them to their final cellular destinations.
Q.23	(B)	Smooth ER is agranular (lacks ribosomes) and specializes in lipid/steroid biosynthesis and calcium storage/detoxification.
Q.24	(B)	Exocytosis is the process wherein intracellular secretory vesicles fuse with the plasma membrane to discharge their contents into extracellular space.
Q.25	(B)	Autophagy is a lysosomal degradation pathway where damaged cellular organelles and cytoplasmic components are engulfed and digested.